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Image Analysis Report

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Source	corrosion.jpg
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Executive Summary

This assessment identifies severe (Level 4) localized corrosion with significant scaling and delamination of the protective coating. The corrosion is concentrated along a horizontal seam or weld, indicating preferential attack or crevice corrosion that may compromise structural integrity.

Key Findings

#	Finding
1	Heavy scaling and localized metal loss in the central region, approximately 30-40% of the visible frame width.
2	Corrosion is highly concentrated along a horizontal feature, likely a weld seam or lap joint.
3	Complete coating failure with visible delamination and exposing the substrate to atmospheric moisture.
4	Extensive rust bleed and staining trailing vertically downward from the primary corrosion sites.
5	Presence of loose, flaky iron oxide (scale) suggesting active, advanced corrosion cells.
6	Exact depth of pitting and remaining wall thickness cannot be determined from this image alone without ultrasonic testing (UT).
7	Visible surface roughness indicates significant material degradation beneath the scale.

Localized detections (VLM, approximate)

- Heavy Scaling (conf. 92%): x=0.350, y=0.120, w=0.380, h=0.650
- Seam Corrosion (conf. 88%): x=0.000, y=0.440, w=1.000, h=0.120
- Coating Failure (conf. 85%): x=0.040, y=0.400, w=0.280, h=0.550

Recommendations

#	Recommendation
1	Perform immediate Ultrasonic Thickness (UT) gauging on a 50mm x 50mm grid across the most affected central area.
2	Execute abrasive blasting to SA 2.5 (Near-White Metal) to remove all loose scale and assess the true extent of metal loss.
3	Conduct Pit Depth measurement using a needle gauge to quantify localized thinning against the original design thickness.
4	If remaining wall thickness is below the minimum allowable (T-min) per API 510/570, perform a weld overlay or install a structural patch plate.
5	Apply a high-build epoxy coating system with a zinc-rich primer once surface preparation is complete and structural integrity is confirmed.
6	Investigate the horizontal seam for crevice corrosion; if it is a lap joint, consider seal-welding if appropriate for the service.
7	Establish a 6-month monitoring frequency for this location until repairs are completed and verified.